

Lab 1

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Section A

```
murder <- c(15.1, 11.3, 7.8, 10.1, 10.3, 6.8, 3.1, 6.2, 10.7, 13.9)
illiteracy <- c(2.1, 1.5, 1.8, 1.9, 1.1, 0.7, 1.1, 0.9, 1.3, 2.0)
```

Part 5

The estimate of β_0 is $b_0 = 1.982$

The estimate of β_1 is $b_1 = 5.242$

The estimated standard error b_1 is $s\{b_1\} = 1.828$

The observed value of the test statistic for the t test of

$$H_0 : \beta_1 = 0 \text{ vs } H_a : \beta_1 \neq 0$$

is $t = 2.867$

The p-value = .02

Is the observed b_1 statistically significantly different from zero (Yes/No)?

Yes

What does this indicate about whether there's any relationship between the murder rate and the illiteracy rate?

There appears to be a positive relationship between murder and illiteracy rate.

$$R^2 = .51$$

Based on the value of R^2 , what percentage of the variation in murder rates is explained by illiteracy rate?

51%

Part 6

Does the assumption of normally distributed errors appear to be met?

Yes.

Part 7

Does the assumption of a constant standard deviation assumption appear to be met?

Yes.

Part 8

a

$$SSR = 60$$

$$SSE = 58.413$$

$$MSR = 60.008$$

$$MSE = 7.302$$

$$SSTO = 118.421$$

$$F = 8.2183$$

b

Give the degrees of freedom for SSR: 1

Give the degrees of freedom for SSSE: 8

Give the degrees of freedom for SSTO: 9

c

Show that F is equal to the t test statistic from step 5.

```
2.867^2
```

```
[1] 8.219689
```

Both p-values are the same, .0209 and .0209.

```
sse <- 58.413
ssto <- 118.421
1 - (sse / ssto)
```

```
[1] 0.5067344
```

.5067 is approximately .51, which, as I rounded, are essentially the same.

Section B

Part 2

The estimate of β_0 is $b_0 = 9.530$

The estimated standard error b_1 is $s\{b_0\} = 1.147$

As no R^2 value shows up here it is 0. As such the R^2 is larger for the model that includes illiteracy as a predictor.

Part 3

The line is horizontal. This makes sense as the predictor of 1 assumes that the rate of murder is constant, unchanging.

Part 4

```
mean(murder)
```

```
[1] 9.53
```

9.53 is the same as the estimate for b_0 .

Part 5

Are the results the same as the test for μ ? Yes.

Section C

Part 2

The interval for β_1 is (1.02, 9.46). This means that the slope could be anywhere between 1.02 and 9.46.

Part 3

The value of the estimate of the mean response is 10.367

Part 4

```
predict(my.reg, newdata = new_dat, interval = "confidence")
```

```
      fit      lwr      upr  
1 10.36868 8.285919 12.45144
```

This says that the mean murder of the new data could be anywhere between 8.26 and 12.45.

Part 5

```
predict(my.reg, newdata = new_dat, interval = "prediction")
```

```
      fit      lwr      upr  
1 10.36868  3.798624 16.93874
```

This predicts that the murder rate of the new data is between 3.79 and 12.94